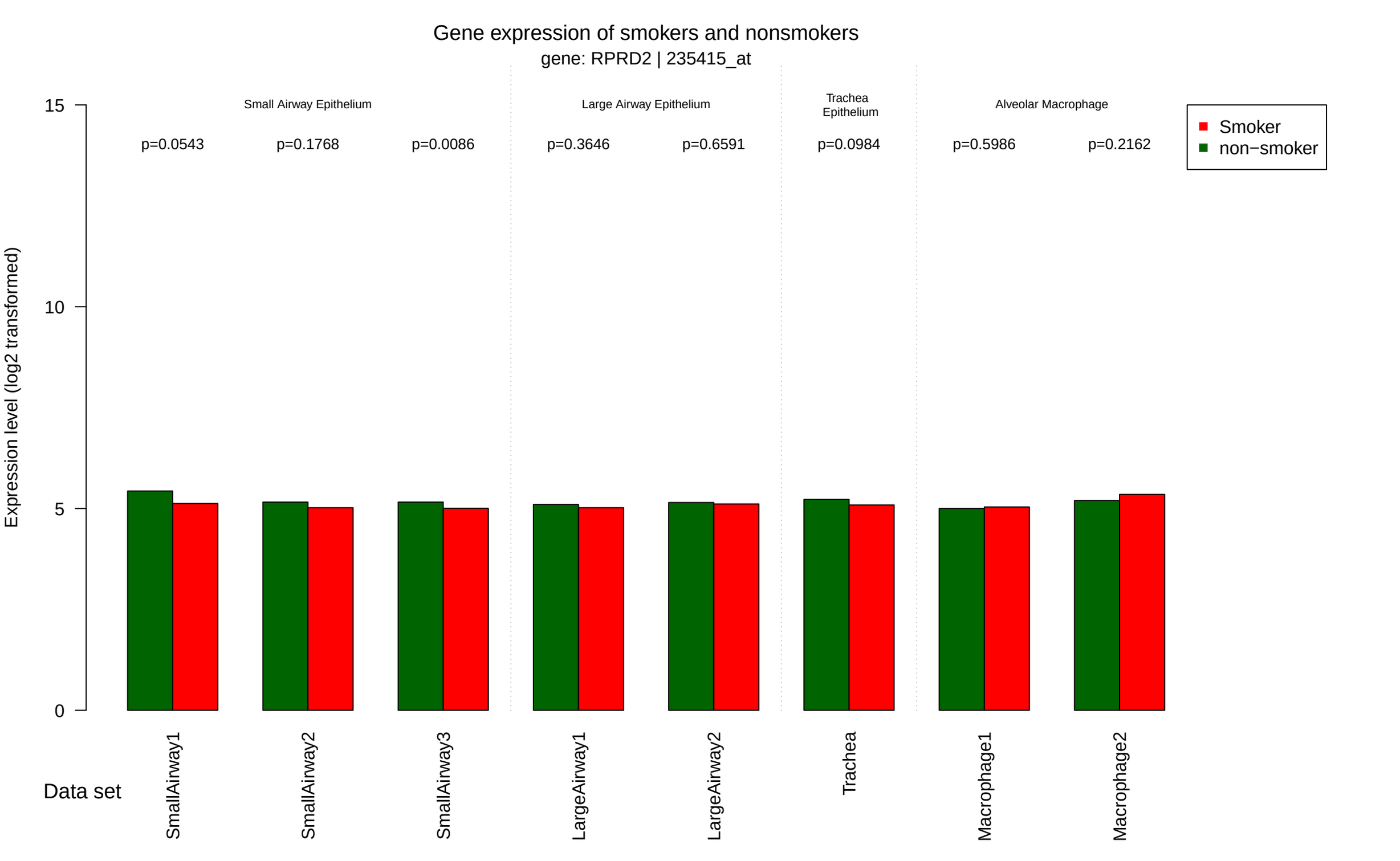
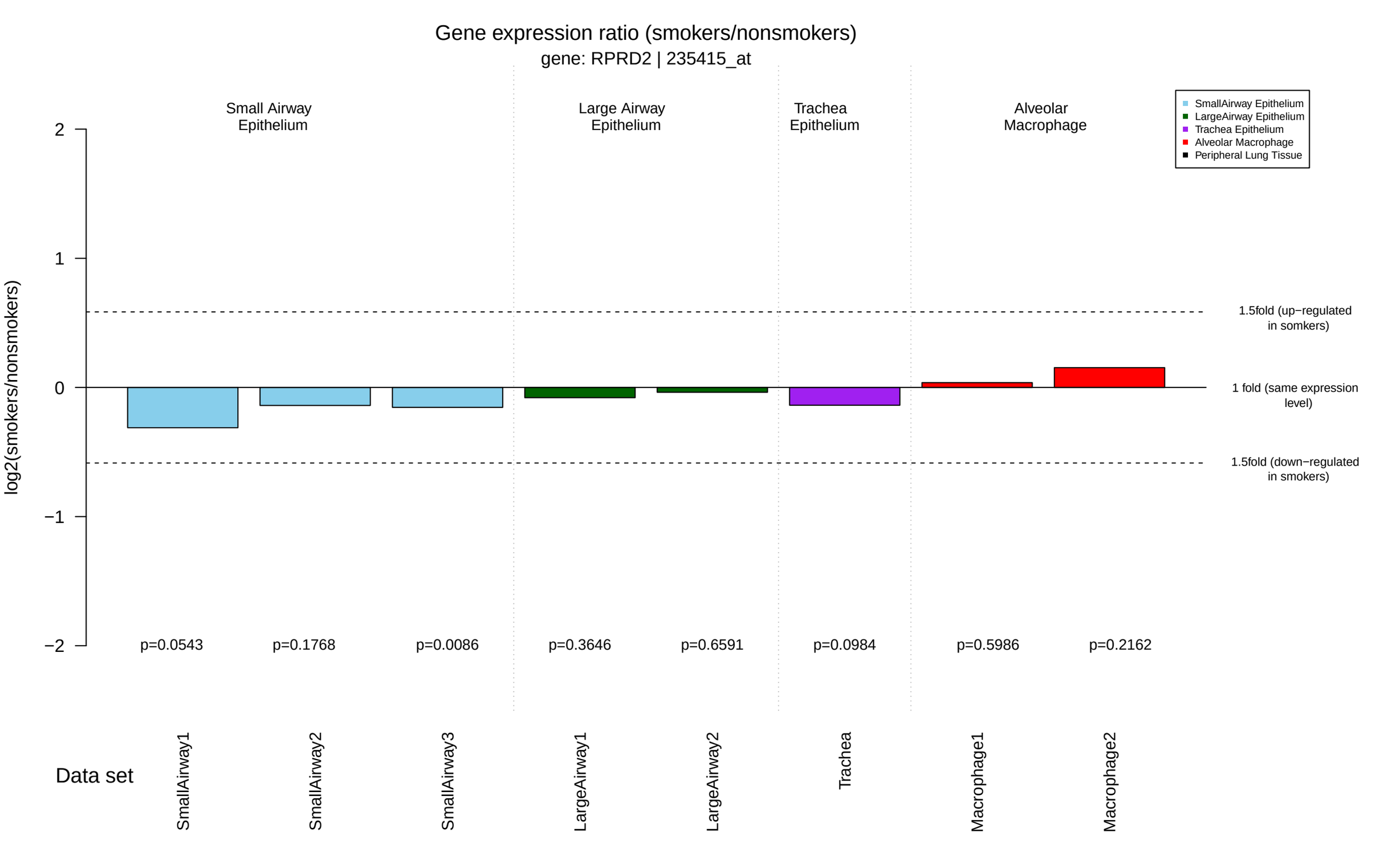
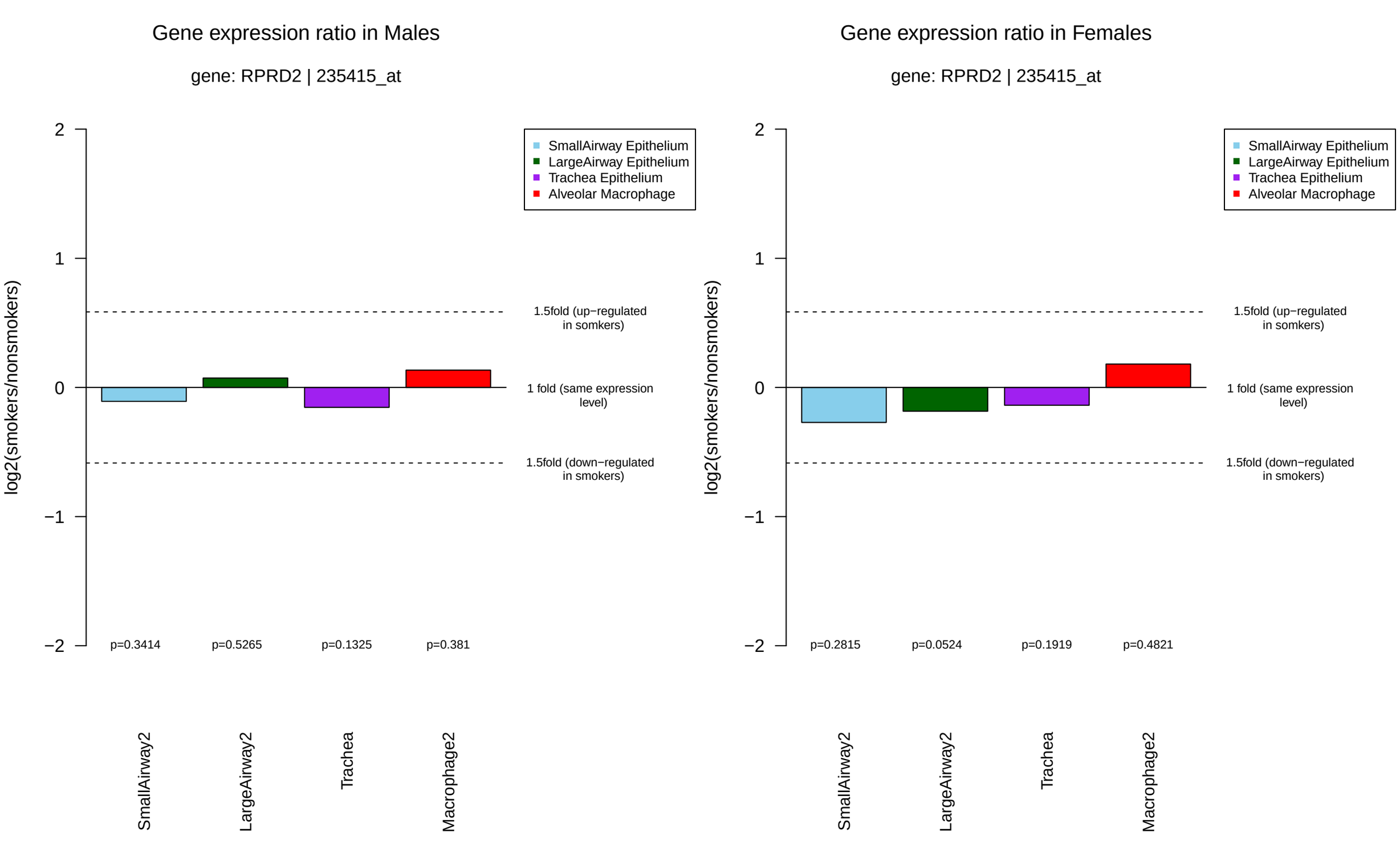
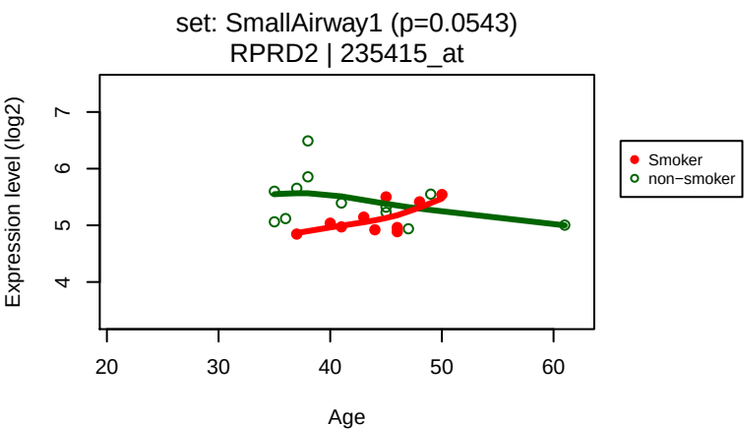
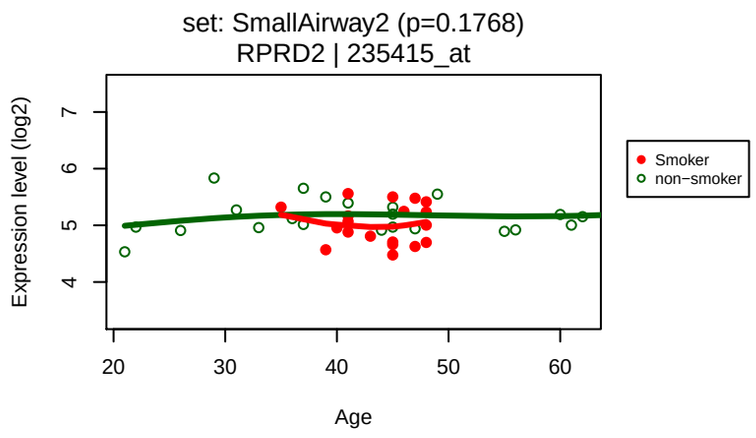
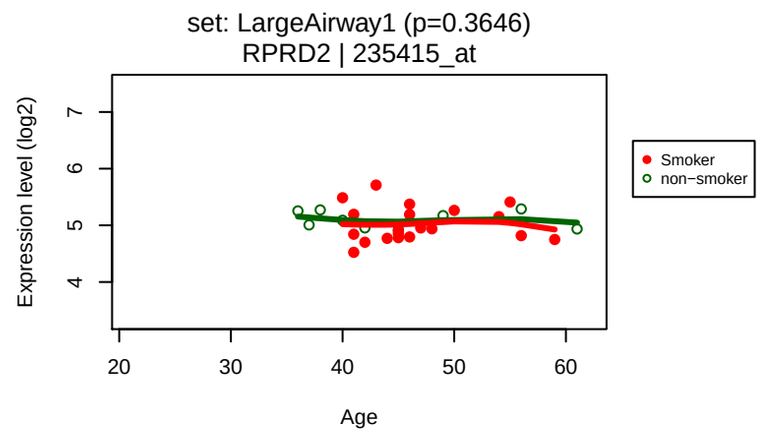

Gene expression of smokers and nonsmokers
gene: RPRD2 | 235415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0543
p=0.1768
p=0.0086
p=0.3646
p=0.6591
p=0.0984
p=0.5986
p=0.2162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

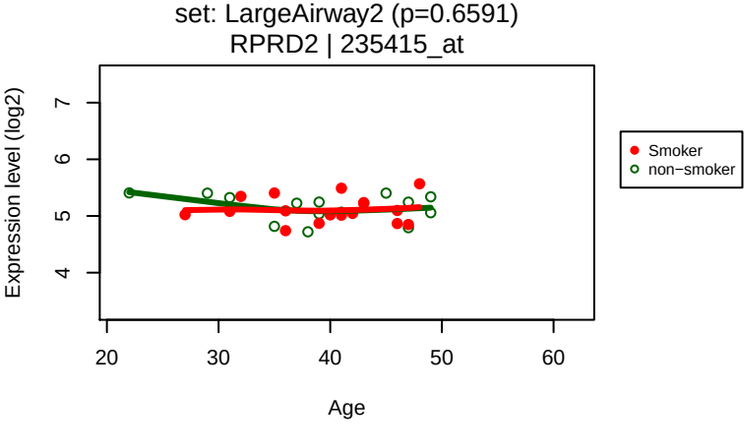

Gene expression ratio (smokers/nonsmokers)
gene: RPRD2 | 235415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0543
p=0.1768
p=0.0086
p=0.3646
p=0.6591
p=0.0984
p=0.5986
p=0.2162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

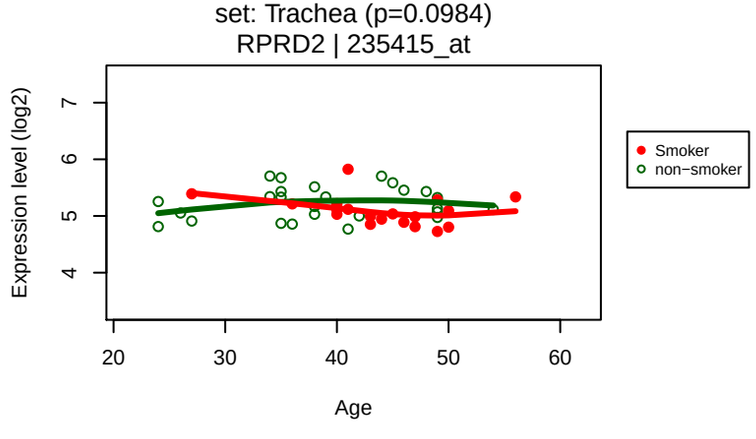

Gene expression ratio in Males
gene: RPRD2 | 235415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3414
p=0.5265
p=0.1325
p=0.381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD2 | 235415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2815
p=0.0524
p=0.1919
p=0.4821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

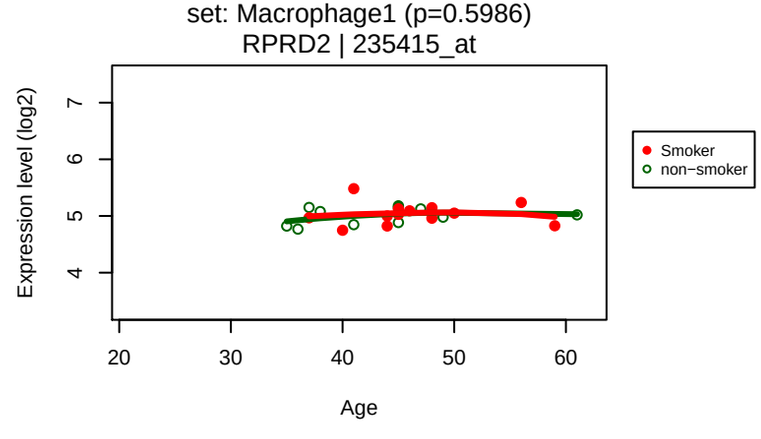

set: SmallAirway1 (p=0.0543)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker

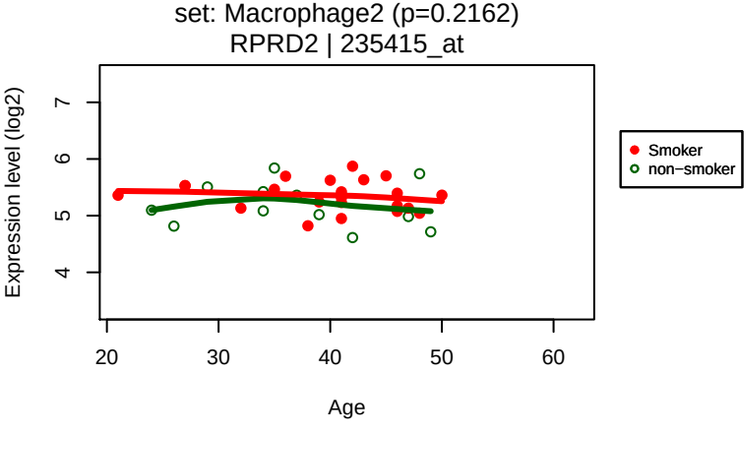

set: SmallAirway2 (p=0.1768)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker

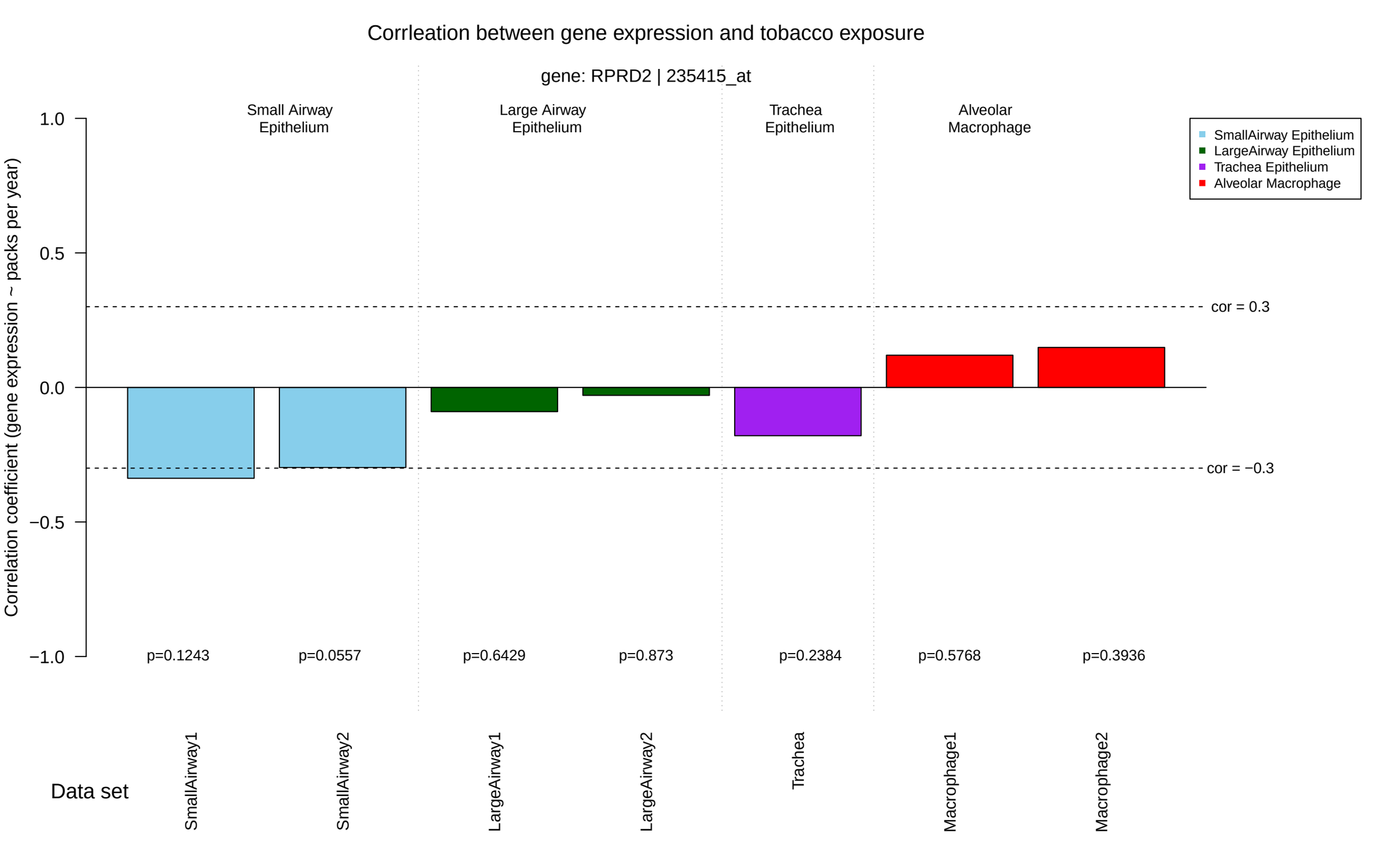

set: LargeAirway1 (p=0.3646)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6591)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0984)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5986)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2162)
RPRD2 | 235415_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD2 | 235415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1243
p=0.0557
p=0.6429
p=0.873
p=0.2384
p=0.5768
p=0.3936
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2